



# Controlling the Pre-bending Delay During Laser Sheet Metal Forming Under Different Atmospheres

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## Abstract

Laser forming is a contactless manufacturing method that shows great promise in revolutionizing in-space manufacturing. However, little is known about the role that different environments have on the bending process. This paper seeks to begin to answer this problem by investigating the influence of different atmospheres on the bending process: atmospheric pressure air, low pressure air, and atmospheric pressure argon. An analysis of the bend angle and temperature during the bending process for the different environments identified a delay between initial metal–laser interaction and the initiation of bending. The duration of the delay is controlled by the laser parameters, the material condition, and the surrounding environment with higher laser powers, softer materials, and materials with thicker initial oxide layers having a shorter pre-bend delay. It was also observed that atmospheres with less oxygen result in lower temperature and thus smaller bend angles than higher-oxygen atmospheres but that creating an oxide layer on the surface of the sample before bending can increase the temperature during bending and the final bend angle. An understanding of how the lack of oxygen influences the bending process is crucial to adapting the laser forming process to the vacuum of space.

**Keywords** Laser forming · Laser bending · Laser ablation · Oxidation · Aluminum · Vacuum

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Extended author information available on the last page of the article

## Introduction

The thermal gradients induced by rastering a laser across a sheet metal surface can cause plastic deformation, bending the sheet. This effect has been developed into the technology of laser forming, a precise and contactless sheet metal forming process. While originally developed for shipbuilding and recently used primarily for prototyping, there is tremendous promise for laser forming to be used for in-space manufacturing. With simple feedstock that is easy to pack and send to orbit and without the need for specific tooling or dies, laser forming seems ideally suited for space applications. However, the effects of the space environment—particularly the lack of atmosphere—require investigation.

During laser forming, the moving laser heats a surface of a sheet, which causes a gradient in temperature and thermal expansion through the thickness [1, 2]. In the thermal gradient mechanism of laser forming, the cooler surrounding material constrains further thermal expansion of the hotter region. As the temperature increases, the force of this constraint surpasses the decreasing temperature dependent yield strength of the thermally expanded material. At this point, the softer thermally expanded material plastically deforms, creating a driving force for bending. Upon cooling, the thermally expanded volume contracts, resulting in net bend about the laser line [3, 4]. There exists a threshold energy below which the temperature in the material is insufficient to induce enough thermal expansion or plastic deformation for bending to occur [5]. Increasing the number of passes can serve as a method of increasing the energy accumulated into the sheet to reach the threshold energy and end the pre-bending delay before linear bending occurs. Additionally increasing the laser power for an individual pass can also serve to achieve the threshold energy with a smaller number of passes [6, 7].

The environment in which laser forming is performed influences the bending efficiency by modifying the cooling rate [8, 9] and absorptivity of the sample to the laser [10]. Adding forced external cooling both increases the final bend angle and decreases the duration of the cool down time between each laser pass [11]. Preferentially cooling the bottom of the sample increases the thermal gradient, encouraging bending [8, 12] while only cooling the top of the sample can reduce the temperature gradient and reduce the rate of bending [10, 11]. The thermal conductivity of the gas being used to cool the sample influences the thermal gradient: helium has a higher thermal conductivity than air and thus cools the sample more and results in a larger bend angle [10]. Additionally, using an inert shielding gas also inhibits the formation of a surface oxide layer that increases the absorptivity to the laser and increases bend angle [10, 13]. The formation of an oxide layer or melt pool during the first pass of the laser forming process increases the absorptivity and the bend angle per scan for the second laser pass [14].

In the vacuum of space, there is no additional external cooling and no oxygen present for oxidation. However, the only study that has investigated laser bending in a vacuum environment used the pressure differential from applying a vacuum to the opposite side of the sample from the laser [15]. This induced an extra stress on the bend line and created complex geometries at lower laser powers and with fewer laser

passes than would otherwise be required [15]. To the authors' knowledge, no studies have been conducted on the efficacy of laser forming in a vacuum environment.

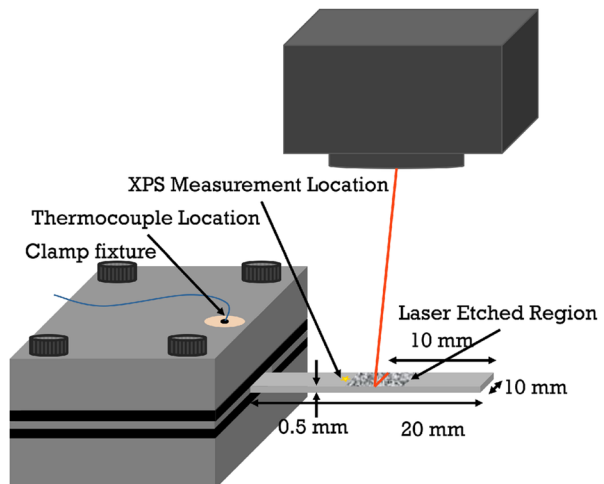
This study deconvolves the oxidation and convective cooling effects of the atmosphere on laser bending. This is achieved through bending trials in varied atmospheres (air, low pressure air, vacuum, and argon), and using specimens heat treated to develop different oxidation layers prior to bending.

## Methods

Rolled AA 3003 aluminum sheets, received from McMaster-Carr in the H14 temper condition, were cut using a Keyence MD-F5200 1090 nm 50 W marking laser to dimensions of 10 mm x 40 mm x 0.5 mm, where the length aligns to the rolling direction. This laser was used throughout this work to perform the laser engraving and laser bending steps. Half of the samples were then heat treated in a Barnstead Thermolyne Type 47900 benchtop furnace at ambient air pressure for 30 minutes at 450°C to identify the influence of material properties and surface oxide layer thickness. For laser engraving and bending, one end of the sample was then placed in an insulated clamp surrounded by 3 mm thick Viton rubber on top and bottom such that the sample extended 20 mm out of the clamping arrangement. This setup is depicted in Fig. 1. For samples bent in low pressure, vacuum, or argon, this sample clamping setup was then placed inside of an Ideal Vacuum Products 6x6x6 cubic modular vacuum chamber with a fused quartz laser window. This laser window has a total transmissivity of 95% for the wavelength of the laser.

Before the samples were bent, they were laser engraved in a 10 mm by 10 mm square using 40 W, 3600 mm/s, and a 60  $\mu\text{m}$  spot size centered along the future laser line to uniformly roughen the surface and increase the absorptivity of the sample. This laser engraving step increased the absorptivity from 2% to 6% as measured at room temperature using a tunable NKT Photonics SuperK EXTREME white laser and a Photon etc. GRAND-EOS hyperspectral imager in an integrating sphere setup.

**Fig. 1** Schematic of sample clamping and lasing fixture. Location of XPS measurement location is shown just outside of the laser engraved region



There was a 40 second pause between the laser engraving step and bending to allow the sample to completely cool down and avoid any latent heat build-up prior to bending. All the laser power and speed conditions, as well as the atmospheric and material conditions used in this paper are delineated in Table 1. The full data set with number of replicates and average bend angle can be found in the [Supplementary Materials](#). The number of passes for each power and speed condition is different to keep the total energy input into the sample the same, as described later in this Section. All samples were bent with a circular Gaussian spot size of 0.44 mm and a pulse rate of 60 kHz for an average pulse on and off time of 220 ns and 16.44  $\mu$ s, respectively. Unless otherwise specified, samples are bent according to the first row of Table 1.

Every sample was bent using the laser along the 10 mm axis of the top surface at a distance of 10 mm from the free end. The inter-pass cool down delay was 0.2 s for 50 mm/s scan speeds and 0.4 s for the 25 mm/s scan speeds, as described in the final column of Table 1. The laser always scanned in the same direction so that each point along the scan line cooled for the same amount of time. This was designed to allow sufficient cooling between each pass to minimize the thermal buildup at the edges of the sample, which can damage the sample through melting or vaporization, while still allowing for sufficient thermal gradient buildup for bending [12, 16].

The temperature was measured on representative samples for each condition using a type K thermocouple that was inserted into a hole drilled in the sample mount that was situated along the sample center line 7.5 mm from the unbent end of the sample and 22.5 mm from the laser line, as illustrated in Fig. 1.

The impact of heat treatment time and temperature on bending was investigated using the heat treatment conditions laid out in Table 2 after which the samples were air cooled. These samples were bent according to the first row of Table 1, with a power of 50 W and laser scan speed of 50 mm/s. The hardness was measured after heat treatment using a Vickers micro-hardness indenter at 100 gf.

To separate the effects of material softening, formation of an ablation channel, or thickening of the oxide layer from the rising temperature increasing the difference between the thermal-stress and the decreasing yield stress, certain experiments had a cool-down pause in the middle of the laser forming process. For these samples, the laser was scanned until an energy of 596 J/mm<sup>2</sup>, pass number 150, was reached, Table 1, followed by a 5 minute pause in which the laser did not mark the sample,

**Table 1** Laser, atmospheric, and material condition parameters varied for all samples in this paper. The number of passes differs for each parameter set to keep the total energy input to the sample the same

| Power (W) | Speed (mm/s) | Air pressure (Torr)       | Temper condition  | Number of passes | Total energy input (J/mm <sup>2</sup> ) | Inter-pass Delay (s) |
|-----------|--------------|---------------------------|-------------------|------------------|-----------------------------------------|----------------------|
| 50        | 50           | 760, 7.6, 0.76, 760 Argon | H14, 30 min 450°C | 300              | 1193                                    | 0.2                  |
| 50        | 25           | 760                       | H14, 30 min 450°C | 150              | 1193                                    | 0.4                  |
| 40        | 50           | 760                       | H14, 30 min 450°C | 376              | 1196                                    | 0.2                  |
| 40        | 25           | 760                       | H14, 30 min 450°C | 188              | 1196                                    | 0.4                  |
| 30        | 50           | 760                       | H14, 30 min 450°C | 500              | 1193                                    | 0.2                  |
| 30        | 25           | 760                       | H14, 30 min 450°C | 250              | 1193                                    | 0.4                  |

**Table 2** Heat treatment time and temperature to ascertain the effect that decreasing the hardness and increasing the thickness of oxide layer had on the pre-bend delay and final bend angle

| Temperature (°C) | Time (min)                        |
|------------------|-----------------------------------|
| 300              | 30, 60                            |
| 350              | 30, 60                            |
| 375              | 30                                |
| 400              | 30, 60                            |
| 415              | 30                                |
| 450              | 5, 15, 30, 60, 120, 180, 300, 480 |
| 500              | 30, 60                            |
| 550              | 30                                |

followed by a repetition of the laser scanning process for another  $596 \text{ J/mm}^2$ . Some samples had two cool-down pauses in which the the total input energy for each stage was decreased from  $596 \text{ J/mm}^2$  to  $397 \text{ J/mm}^2$ , after 100 laser passes elapsed, to keep the total input energy consistent at  $1193 \text{ J/mm}^2$ .

The bends were recorded using a Panasonic Lumix digital camera that took a side-on video. The video was then processed using an open-source MATLAB tool developed by the authors (called NINJA, which is available at [github.com/MONSTERgroup/NINJA](https://github.com/MONSTERgroup/NINJA)) which uses image processing techniques to automatically calculate the bend angle for each frame of the video. In short, the video is cropped to solely the bending region, then each frame is converted to grayscale and the contrast is increased between the specimen and the background. A radon transform is applied to the frame, which projects lines in an  $(x, y)$  space to points in an  $(x', \theta)$  space, where  $x'$  is the offset of the line from an origin, and  $\theta$  is an angle analogous to slope of the line. A peakfinding algorithm finds the strongest peaks in the transformed map, which belong to line segments within the specimen. One set of peaks remains constant (the fixed region of the specimen) and the other set of peaks change position in  $\theta$  (the free region of the specimen). For each frame, the different in theta between these average positions of the sets of peaks is the bend angle. From these videos, the ablation depth on the laser exit side was measured manually and compared against the bend angle for specific frames.

The thickness of the oxide layers that developed during the laser forming process were measured after bending using an ULVAC-PHI x-ray Versaprobe that uses an Al K- $\alpha$  x-ray source to measure the peak broadening of the Al2p peak. For all samples, a brief Ar sputter step was conducted before XPS measurement to remove the surface contaminants and the adventitious carbon layer. For some conditions, the oxide thickness was greater than the x-ray penetration depth of about 10 nm, so subsequent argon sputter steps were done to reach the full extent of the oxide thickness. For these conditions, the amount of material removed during the sputter was added to the measured thickness of the oxide layer. The XPS measurements were done on the bulk material 5 mm from the laser line, just outside of the laser engraved region, as shown in Fig. 1. The relative intensity of the Al metal peak and the Al-O peak can be adapted to a thickness value of the oxide using the equation proposed by Strohmeir [17, 18].

The bend profiles were calculated in terms of bend angle per accumulated energy absorbed, or the total energy transferred from the laser into the sample over the course of the bending process. Using the accumulated energy density (AED) enables a more consistent comparison between bend experiments that had different laser power and

speed parameters, taking into account both the number of passes and the length of time for each pass [19].

Ramos et al. proposed (1)

$$AED = NP/Dv \quad (1)$$

where  $N$  is the number of laser passes,  $P$  is the laser power in watts,  $D$  is the beam diameter in mm, and  $v$  is the scan speed in mm/s [20]. The Ramos Equation, Eq. 1, assumes complete absorption of the laser by the sample, but it is known that metal is highly reflective, especially to the wavelengths commonly used by lasers [21]. To account for this, Shun-Hong et al. modified the Ramos AED formula into Eq. 2 to also include an absorptivity factor  $\eta$  [22].

$$AED = \eta NP/Dv \quad (2)$$

Equations 1 and 2 can only account for the total energy absorbed at the end of every pass. The processed video data generated in this paper provides a measured bend angle 30 times per second, several times per each individual laser pass. To fully exploit the data, Eq. 3 was derived to account for energy input for elapsed time and is independent of laser scanning speed.

$$AED = tP\eta/A \quad (3)$$

In this equation,  $t$  is the time in seconds that the laser has marked the surface of the sample. The new equation also uses the area  $A$  in mm<sup>2</sup> of the laser beam to keep the units consistent.

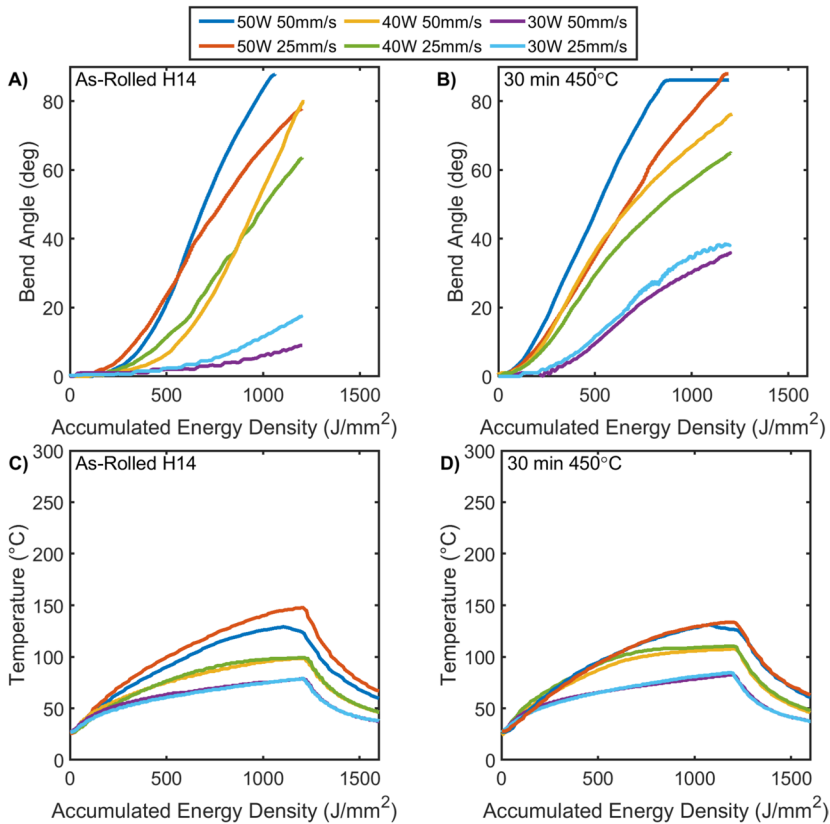
$$D = (4F\lambda M^2/\pi D_L) * \sqrt{1 + (\pi(L - F)/\lambda M^2(2F/D_L)^2)^2} \quad (4)$$

The area was calculated using the beam diameter equation provided by Saleh et al. where  $F$  is the focal length of the laser,  $\lambda$  is the wavelength,  $M^2$  is the beam quality factor,  $D_L$  is the beam diameter at the lens, and  $L$  is the distance from the lens [23]. This provides a measurement of the energy accumulation in the sample over time.

## Results

### Impact of Laser Parameters on Bend Profile

Representative bend curves are seen in Fig. 2A for a variety of power and speed combinations. Notably, there is a period of time at the beginning of the bend curve during which the rate of bending is very low, which will be referred to as the pre-bending regime. As time and the total energy input progresses, the rate of bending increases in an exponential fashion until it reaches a constant slope, the linear bending regime. The extent of the pre-bend delay was determined by identifying the linear region of

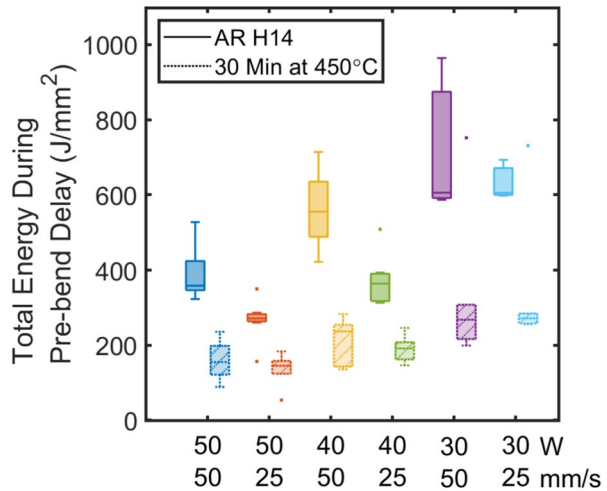


**Fig. 2** Increasing the power input decreases the length of the pre-bend delay. **A)** shows the as-rolled condition while **B)** shows bending after a 30 minute heat treatment at 450°C. Performing the heat treatment also decreases the length of the pre-bend delay and increases the final bend angle. Thermocouple measured temperatures for all power and speed conditions. Temperature buildup and maximum temperature reached is consistent for samples with the same power regardless of scan speed

the bend, the portion of the bend profile that fit a linear regression with the highest  $r^2$  value, and calculating where that linear line intercepted the x-axis. The x-intercept was then taken to be the dividing line between the pre-bending regime and the linear regime.

Increasing the power of the laser decreases the duration of the pre-bend delay and increases the final bend angle. As seen in Figs. 2 and 3, regardless of the laser speed or the heat treatment condition, the higher power laser scan has a shorter pre-bend delay and a lower total required energy input to reach the same bend angle than the lower power laser scans. In the as-rolled H14 condition, Fig. 2A and the solid box-plots in Fig. 3, the lowest power (30 W) and fastest speed (50 mm/s) laser conditions have the longest pre-bend delay and the lowest final bend angle of any condition. The two conditions at 50 W have similar pre-bend delays and matching bend curves until about 600 J/mm<sup>2</sup> input energy, at which point the rate of bending decreases for the

**Fig. 3** The total energy accumulated during the pre-bend regime, after which bending moves into the linear regime. For all power and speed conditions the pre-bend regime is shorter after a heat treatment, with a more significant decrease occurring in the lower power samples. After heat treatment, there is little variation in pre-bend regime across the power and speed combinations but in the as-rolled states, both of the 30 W samples and the 40 W 50 mm/s sample had a much longer pre-bend delay than the 50 W or 40 W 25 mm/s samples



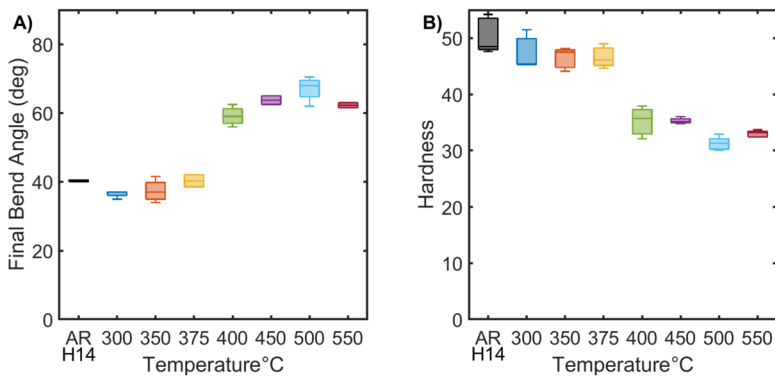
25 mm/s samples and remains constant for the 50 mm/s samples. The sample bent at 40 W and 50 mm/s has a similar pre-bend delay as both of the 30 W samples, but has a higher bend per energy slope during the linear regime, enabling it to reach the same bend angle as the 50 W and the other 40 W samples.

Heat treating at 450°C for 30 minutes before laser forming, Fig. 2B and the dotted box-plots in Fig. 3, shortens the pre-bend delay and increases the final bend angle for all conditions. After heat treating, all conditions have effectively the same duration of the pre-bend delay, although it is slightly longer for the 30 W samples. The 50 W 50 mm/s sample still reaches the largest bend angle, but does so with less input energy required than the as-rolled samples. The 40 W 50 mm/s samples has the largest difference in pre-bend delay after heat treatment. Without the heat treatment, the 40 W 50 mm/s samples had the 4<sup>th</sup> longest pre-bend delay but a similar slope to the 50 W 50 mm/s samples, resulting in the same final bend angle as the 50 W 25 mm/s samples. After the heat treatment, the 40 W 50 mm/s samples have a similar bend profile to the 50 W 25 mm/s samples, with a significant decrease in the slope at 600 J/mm². The 30 W samples now closely match each other exhibiting little difference in pre-bend delay duration or final bend angle for the different speeds.

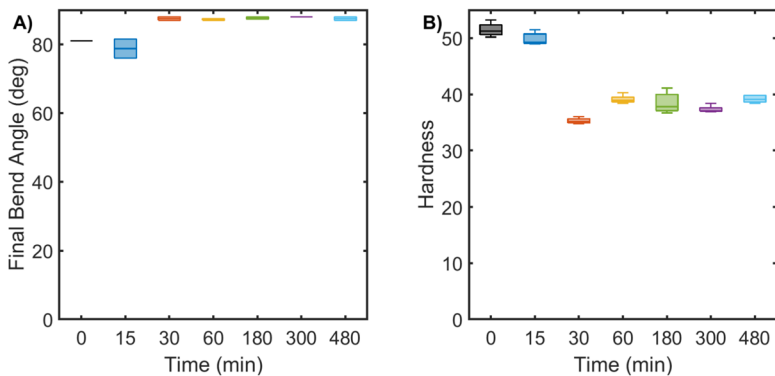
The temperature was measured using a thermocouple placed on top of the sample 25 mm away from the laser line (Fig. 1). The maximum temperature reached and the temperature per energy profile depends on the laser power, not the scan speed, as shown in Fig. 2C and D. The average temperatures are similar for the as-rolled and heat treated samples, not displaying any significant deviation in maximum temperature after heat treating. For the heat treated samples, Fig. 2D, all the samples have similar temperature build up rates before each power condition plateaued at successively higher temperatures.

### Impact of Heat Treatment Condition on Bend Profile

Performing a heat treatment above 400°C increases the final bend angle, as seen in Fig. 4A. The length of time at temperature does not significantly impact the bend



**Fig. 4** The final bend angle after  $600 \text{ J/mm}^2$  for variations in heat treatment temperature for a 30 minute heat treatment. **B)** shows the associated Vickers microhardness (HV) after heat treatment but before bending

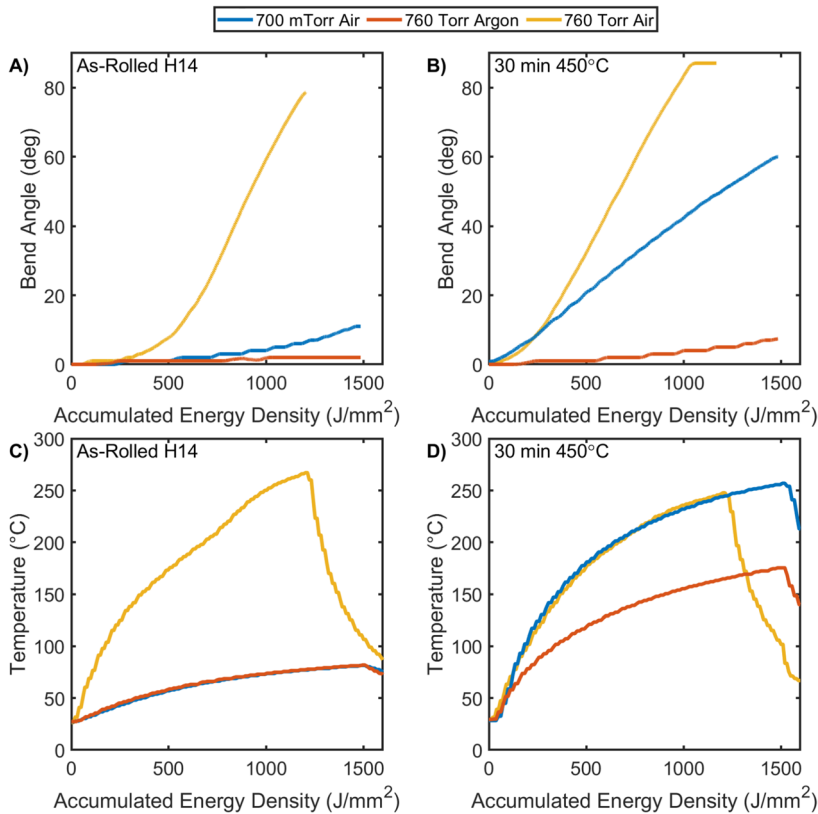


**Fig. 5** The final bend angle after  $1200 \text{ J/mm}^2$  for different durations of heat treatment at  $450^\circ\text{C}$ . **B)** shows the associated Vickers microhardness (HV) after heat treatment but before bending

angle; heat treating at  $450^\circ\text{C}$  for longer than 30 minutes and less than 8 hours had effectively the same bend angle, Fig. 5A. The Vickers hardness values for the heat treatments shown in Fig. 4 are split into two groups: those less than  $400^\circ\text{C}$  and those heat treated at temperatures greater than or equal to  $400^\circ\text{C}$ , with the higher heat treatment temperatures being softer. Increasing the duration of heat treatment softens the material until 30 minutes have elapsed at which point the hardness values plateau. Additionally, an increase in the oxide thickness with heat treatment temperature and time was observed, the results of which will be elaborated on in “[Oxide Layer Thickness After Bending](#)” section.

### Impact of Surrounding Environment on Bend Profile

Figure 6A shows the bend profile after bending as-rolled samples in different atmospheres: 760 Torr air, 760 mTorr air, and 760 Torr argon. Bending in argon had the

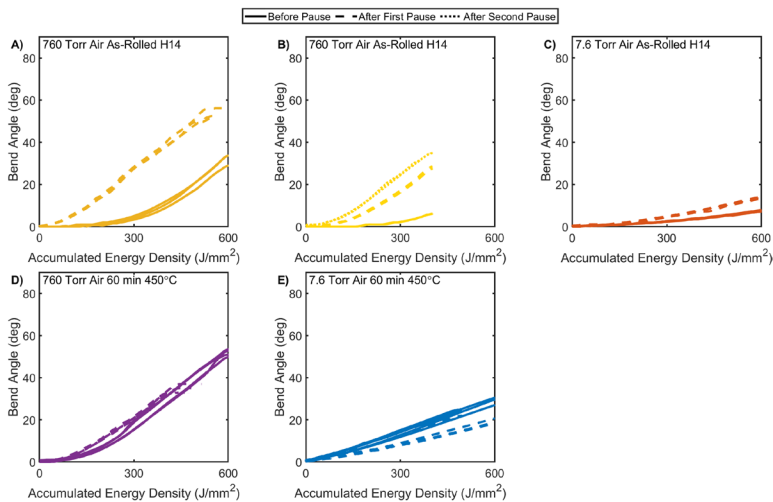
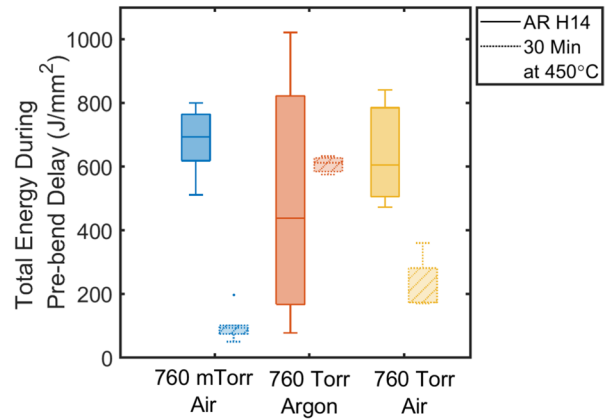


**Fig. 6** Bend and temperature profiles for bending in different atmospheres with different heat treatment conditions. **A)** and **B)** show the bend profiles while **C)** and **D)** show the temperature profiles. Samples in **A)** and **C)** are in the as-rolled H14 condition while samples in **B)** and **D)** have been heat treated for 30 minutes at 450°C before bending

lowest bend angle while bending in ambient air pressure had the highest bend angle. Performing a heat treatment before bending, Fig. 6B, results in higher bend angles for all conditions. At 760 Torr air there is little difference between the temperature of the heat treated and as-rolled samples, seen in Fig. 6C and D. The temperature for the as-rolled samples bent at 760 Torr argon and 700 mTorr air had the same temperature profile. Heat treating increased the maximum temperature reached for both 760 Torr argon and 700 mTorr air, with the 700 mTorr air reaching a similar temperature to the 760 Torr air samples and no longer matching temperature of the 760 Torr argon samples. All samples were bent at 50 W and 50 mm/s according to the first row of Table 1. Figures containing the full data set for Figs. 2 and 6, along with the full data set containing the average bend angle and standard deviation along with the number of replicates for each condition can be seen in the Supplementary Materials Online Resource 1.

In Fig. 7, there is little variation in the mean energy accumulated during the pre-bend regime for samples in the as-rolled condition. After a heat treatment the energy accumulated in the pre-bend delay decreased for the 760 mTorr and 760 Torr samples

**Fig. 7** Energy accumulated during the pre-bend regime for samples bent in different atmospheres with different heat treatment conditions



**Fig. 8** Samples bent with a five minute cool down pause before restarting bending. There were two cool down pauses for the samples in **B)** but only one pause for **A), C), D),** and **E)** in different heat treatment and environmental conditions. The lighter shaded bend curves in **B)** denote samples that had cool down pauses after 100 passes instead of the full 150 passes that all the rest had. The pre-bend delay changes for the different environmental conditions, being shortest in **E)** and longest in **C)**. For both **A)** and **C)**, the pre-bend is shorter after the pause while in **E)** the average rate of bending is lower after the pause

but remained the same for the samples bent in argon. The 760 mTorr samples saw the largest decrease in and smallest total energy accumulated during the pre-bend delay after a heat treatment.

### Sample Change During Bending Process

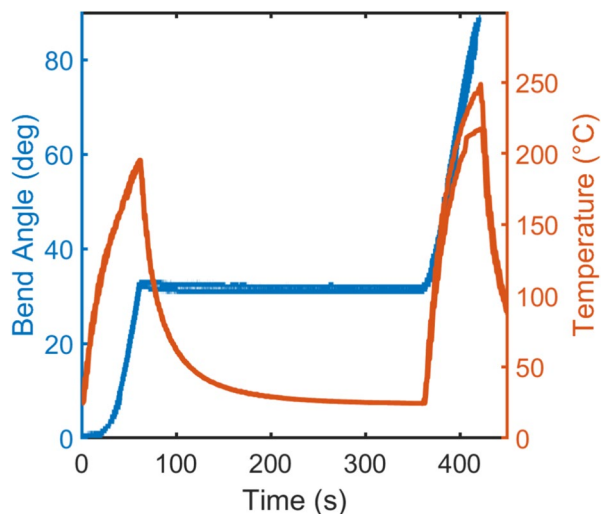
In Fig. 8, the samples were bent with at least one five-minute cool down pause in the middle of the bending process, as described in “Methods” section, in different heat treatment and environmental conditions, as laid out in the first row of Table 1. In the

results shown in Fig. 8, the bending segment after the pause was relocated to the origin to provide more direct comparison of the bend angle and the duration of the pre-bend delay for the segments before and after the pause. Bending at 7.6 Torr in both heat treatment conditions has a lower final bend angle than the corresponding bends at 760 Torr. The samples heat treated for 60 minutes at 450°C and bent at 7.6 Torr have almost no pre-bend delay before the pause, bending with the first laser scan. However, the rate of bending is lower than the samples bent at 760 Torr for the same heat treatment condition. For all samples bent in the as-rolled H14 condition regardless of atmosphere Fig. 8A and C, in the bend segment after the cool down pause the pre-bend delay was shorter and the sample reached a higher final bend angle. Heat treating for 60 minutes at 450°C before bending results in effectively no difference in the rate of bending after the cool down pause, Fig. 8D.

In Fig. 8A and B, there are two sets of bend curves using the same laser power and speed. Both sets were bent with a total energy input of 1193 J/mm<sup>2</sup>, but in Fig. 8B the total energy was split into three segments of 397 J/mm<sup>2</sup> with two five-minute cool down pauses, one between each segment. In Fig. 8A, the total energy input was split into two segments of 596 J/mm<sup>2</sup> with only one five-minute cool down pause. For the two pause curves, the duration of the pre-bend delay decreases and the bend angle increases after the first pause. After the second pause the final bend angle and pre-bend delay duration decreased less than after the first pause. Additionally, the bend curves after the first pause for the samples bent to 397 J/mm<sup>2</sup> do not match the corresponding bend curves for samples bent to 596 J/mm<sup>2</sup>. Instead, the third segment of the bends with 397 J/mm<sup>2</sup> sections best matches the second segment of the bends with 596 J/mm<sup>2</sup> sections.

The increase in temperature of a sample from Fig. 8A differs before and after the five minute cool down pause, seen in Fig. 9. Before the pause the temperature increase slows down as bending progresses and appears to be approaching a maxi-

**Fig. 9** Temperature measurements of bending process compared with bend angle



imum temperature plateau if bending were not paused. After the pause, the temperature increase is steeper and reaches a higher maximum temperature.

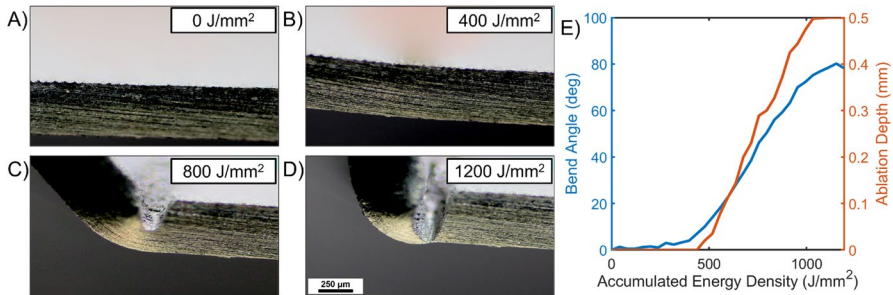
### Laser-Induced Ablation Channel Formation

Similar to bending, laser induced ablation does not occur instantaneously with the initial laser-metal interaction. Figure 10A through D shows frames from the bending video from which the bend angle and ablation depth at the laser exit are measured. Shortly after the initial laser-metal interaction but before bending begins, Fig. 10A, the sample is flat with no ablation present, only minor surface roughness from the laser engraving step described in “Methods” section. By 400 J/mm<sup>2</sup>, Fig. 10B, minor bending has begun but no ablation can be observed. Between 400–800 J/mm<sup>2</sup>, Fig. 10B and C respectively, both bending and ablation have begun. After this point, both bending and ablation continue linearly. This relationship can be seen in Fig. 10E where the timing of the initial formation of the ablation channel and when the linear bending regime starts closely match each other.

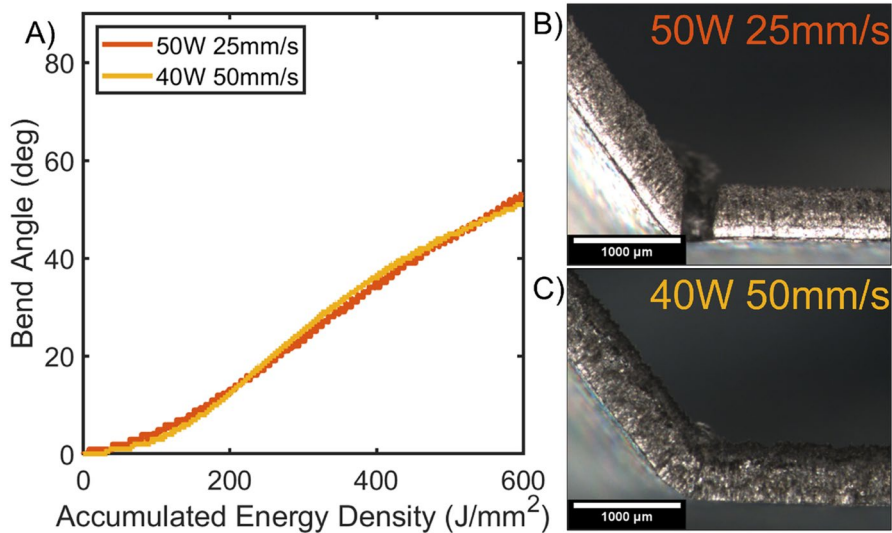
Figure 11A shows two samples with different laser power and speed conditions that have the same total accumulated energy in the sample and nearly identical bend per energy curves. The comparable final bend angles can be seen in Fig. 11B and C in which B shows significant ablation, cutting through the entire thickness of the sample at the exit edge seen in the snapshot. In comparison, C has the same final bend angle and exhibits no ablation at the exit edge.

### Oxide Layer Thickness After Bending

The impact of bending environment on the thickness of the oxide layer after bending is shown in Fig. 12. The unbent specimens have the thin oxide layer typically found on aluminum. A heat treatment at 450°C for 60 minutes slightly increases the thickness of the oxide layer from 6 nm to 8 nm. Bending the as-rolled and heat treated samples in 760 Torr air further increases the oxide layer, with a more significant increase for the heat treated sample than the as-rolled. Bending at 13 mTorr air or 650 Torr argon has little impact on the thickness of the oxide layer. Bending at 7.6 Torr air

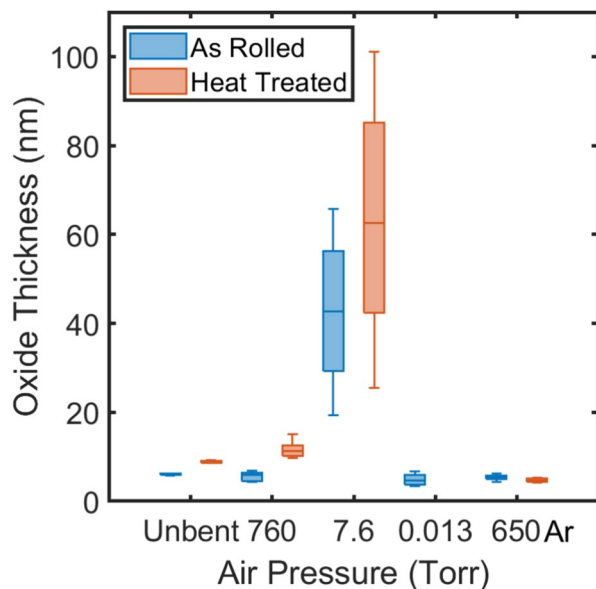


**Fig. 10** The growth of the ablation channel can be seen in **A)** through **D)**, which show side-on snapshots of the bend angle taken every 400 J/mm<sup>2</sup>. **E)** shows that the ablation channel does not begin to form until shortly after pass 100, 400 J/mm<sup>2</sup>, at which point ablation proceeds linearly, and the pre-bending regime ends as the rate of bending picks up



**Fig. 11** Decreasing laser power from 50 W to 40 W and increasing the scanning speed from 25 mm/s to 50 mm/s requires increasing the total number of laser passes from 75 to 188 to maintain the same total energy input, but results in the same bend angle, **A**). Decreasing the laser power decreases the amount of ablation observed on the surface of the sample from 500  $\mu\text{m}$  at the end of the sample in **B**) to 0  $\mu\text{m}$  in **C**)

**Fig. 12** Oxide thickness measured on several samples after bending in a variety of atmospheric and heat-treatment conditions. Half of the samples were heat-treated at 450°C for 60 minutes. The largest increase in the thickness of the oxide layer was during bending at low pressure rather than bending at atmospheric pressure, which had a more modest increase in oxide layer thickness. Bending in medium vacuum or argon atmospheres did not change the thickness of the oxide layer. Samples bent according to the first row of Table 1



has a significant impact on the oxide layer increasing from 6 nm to 42 nm and from 8 nm to 62 nm for the as-rolled and heat treated samples, respectively.

## Discussion

In this study the role of atmosphere on laser bending performance was investigated in the AA3003 Al alloy. Of particular note was the existence of a pre-bending regime, where very little bending was observed during the initial laser passes. McBride et al. proposed a threshold energy analogous to the pre-bending regime, below which no bending will occur [5]. At low energies the temperature gradient is not high enough to cause the critical plastic deformation needed for bending. Increasing the number of passes increases the energy absorbed into the sample to reach the threshold energy. Experiments with a cool down pause in the middle of the bending process revealed a difference in the bending rate and the amount of energy required to reach the threshold energy for bending, indicating that there is an irreversible material change occurring that alters how the material is heating up and reducing the time to reach the threshold temperature for bending. Experiments specifically examining the formation of an ablation channel indicated that these channels cannot explain the existence of a pre-bending delay and do not affect the energy required to reach the threshold bending stress. Experiments on specimens previously subjected to a heat treatment revealed that surface condition plays a substantial role in bending rates and the pre-bending behavior, and XPS measurements illustrated that the surface condition also evolves during bending in air. Comparison of the bend curves between conditions and understanding this surface evolution allows the impact of environment to be ascertained.

### Influence of Temperature Build up on Bend Angle and Pre-bend Delay

The temperature the sample reaches strongly correlates to the final bend angle and the duration of the pre-bend regime. As shown in Fig. 2, samples that reach a lower maximum temperature have a lower final bend angle. Other researchers have also observed that samples that reach a higher temperature during bending have a larger bend angle than those that reach a lower temperature [24, 25]. Samples that reach a higher temperature increase the difference between the thermal-induced stress and the decreasing yield stress, attaining a larger bend angle than samples that reach a lower temperature [5, 26].

The temperature at the measurement location is lower than the temperature at the laser line due to the intervening distance, and so the temperatures reported are only representative of the relative temperatures between conditions and not the actual temperature required for bending. Additionally, the added thermal mass of the thermocouple was sufficient to decrease the bend angle achieved for certain laser parameters. Instead, the temperature collected was compared against the bend per accumulated energy curves of other samples that did not have an attached thermocouple.

It is widely accepted that increasing the line energy ( $P/V$ ) by increasing the laser power ( $P$ ) or decreasing the laser scan speed ( $V$ ) increases the rate of bending and the final bend angle [5, 16]. However, from Fig. 2A, power alone is more predictive of

the bend angle than line energy or scan speed. Increasing the power always increases the maximum temperature, even if the immediate impact on the bend angle or the duration of the pre-bend delay is less immediately obvious. Scan speed does not have a significant impact on the temperature or the bend angle because of the high thermal conductivity and diffusivity of this aluminum alloy. Materials with a high thermal conductivity can, within the interaction time of a single laser pass, create a thermal gradient that imparts small amounts of plastic deformation on the material that thermally expanded and then quickly normalize temperatures and equalize the thermal gradient. For the variation in scan speeds used in this experiment the difference in interaction time is likely insufficient to impact the initial formation of the thermal gradient and the sample has thermally equalized by the time the subsequent laser pass begins. Even though all the samples had the same total energy input, the samples bent to different angles because the total duration of bending was different. The lower power samples required more laser passes and a longer time to achieve the same total energy input as the higher power samples. The smaller energy input was counteracted by the ambient cooling present in the system, reducing the maximum temperature and final bend angle reached.

The sample heats up quickly during the pre-bending regime after which the rate of the temperature rise decreases and the temperature approaches a constant value, as demonstrated in the lower power temperature curves in Fig. 2D. In typical multi-pass laser forming with a short inter-pass delay cooling time, the first laser passes correspond to a period in which the increasing temperature is the primary controlling factor for the bend angle [3]. The rapid increase in temperature causes the bend angle per pass to increase until the sample reaches a steady state temperature, at which point other material factors become dominant. For the higher power conditions, the period of the most rapid temperature increase closely matches the length of the pre-bend delay. This aligns with the theory proposed by McBride et al. in which bending is slow or non-existent until a threshold temperature is reached. At lower laser powers, however, the temperature plateaus well before the pre-bend delay ends. It is likely that at the higher laser powers the thermal energy from the laser is sufficient to create the thermal gradient for bending during the pre-bend delay, but at lower laser powers the ambient cooling hindered the development of the thermal gradient, elongating the duration of the pre-bend delay and decreasing the bend angle. Chan et al. and Yau et al. report a minimum line energy or laser power for bending [6, 7]. At certain laser power and scan speed combinations, there is insufficient energy input over the number of laser passes used in that study to reach the critical temperature and stress for bending. However, it is likely that if the number of passes had been increased, the critical temperature for bending could have been reached for samples bent using a low line energy, mirroring the results seen in Fig. 2A.

Decreasing the cooling rate by bending in different atmospheres should increase the heating rate and the bending rate. The dominant cooling mechanism for bending in both atmospheric pressure argon and air is convective cooling, with radiative cooling only contributing a small amount to the total cooling rate [27]. Decreasing the air pressure below 100 Torr changes the dominant cooling mechanism to radiative cooling, and below 1 Torr, the effect of convection becomes negligible compared to radiative cooling [27]. Additionally, there is not a significant difference between the

convection coefficient of argon and nitrogen at standard pressures [28], so samples in similar pressures of argon and air should cool at the same rate. However, as shown in the as-rolled samples in Figs. 6 and 7, the samples bent in argon reached a lower temperature and bend angle than the samples bent in equivalent pressure of air, while the samples bent in radiation-dominated cooling at low pressures reached a lower temperature than the samples bent in convection dominated cooling at higher pressure.

Heat treating the sample before bending reduces the duration of the pre-bend delay and increases the final bend angle for all conditions, but does not increase the temperature reached when bent at ambient atmospheric pressure. Similar results were found by Wei et al. in which altering the temper condition in stainless steels from as-received to annealed increases the bend angle [29]. Additionally, Guan et al. and Anthony et al. determined that, in general, materials with a lower initial yield strength will bend more [30, 31]. This is demonstrated in Figs. 4 and 5 where samples that softened during the heat treatment reached a higher bend angle and had a shorter pre-bend regime than those that did not. The relative hardness of the material before bending should not play a significant role in the temperature buildup during bending. Despite that, heat treating significantly increased the temperature as well as the bend angle when bending in low pressure air or atmospheric pressure argon environments. Before heat treating, the low pressure air samples reach the same temperature as the atmospheric pressure argon samples, and have similar bend angles. After heat treating, the low pressure air samples reach a comparable temperature and bend angle to the samples bent in atmospheric pressure air. In addition to softening the material, the heat treatment before bending also thickened the oxide layer, and the increased thermal insulation would explain the temperature increase when bending in low oxygen environments.

In 760 Torr air, the sample can readily grow the insulating oxide layer, thus reaching equivalent temperatures for both heat treatment conditions while in the lower pressure or argon environments the thickening of the oxide layer is hindered. The thinner oxide layer retains less heat within the sample leading to a decrease in the thermal gradient and resulting bend angle. The influence of the development of the oxide layer on the bending process will be discussed in more detail later.

Similarly, pausing the bending process and allowing the sample to cool down to room temperature and equalize the thermal gradient increases the final bend angle and maximum temperature when bending resumes after the pause for samples in the as-rolled material condition, best seen in Figs. 8A and 9. It is likely that the heat from the laser during the first stage before the pause causes equivalent thermal annealing and oxidation as the heat treatment prior to bending. Ramos et al. also found that the energy from the laser after only a few passes was sufficient for annealing to occur, but that for certain laser parameters, it was possible to bend with no change in hardness [20]. The heat treated samples bent in atmosphere do not exhibit a significant decrease in duration of the pre-bend delay after the pause, potentially because the softening and oxide layer thickening that occurs during the pre-bend delay when bending as-rolled samples was already achieved by the heat treatment. The pre-bend delay is then simply determined by the time and energy required to achieve the thermal gradient necessary for bending. The bending after the pause in atmosphere of

as-rolled samples matches the bending before the pause of the heat treated samples, further reinforcing this possible explanation.

## **Decoupling the Influence of Surface Features on Pre-bend Delay**

### **Creation of Laser-Induced Ablation Channel**

One possibility for the increasing effect of power on decreasing the duration of the pre-bend delay is that, at the laser fluence of the highest laser power used in this study, an ablation channel formed which increased the absorptivity to the laser. Increasing the energy absorbed increase the temperature the sample reaches and decreasing the time to reach the threshold stress for bending. At higher laser powers, laser ablation creates a channel analogous to the keyhole observed in laser welding or laser powder bed fusion [32]. Increasing the keyhole depth increases the absorptivity of the metal, even accounting for the influence of the surrounding molten material [33]. This increased absorptivity is due to the walls of the ablation channel or keyhole absorbing the diffuse laser reflections that would otherwise be totally reflected off of a smooth surface [21, 34]. Thus, increasing the depth of the ablation channel should have a comparable effect on the absorptivity. However, despite the alignment of the bend angle and ablation channel depth in Fig. 10, the fact that the same bend profile was achieved with no ablation in Fig. 11 suggests that the formation of an ablation channel is a side effect of the laser parameters and does not have a controlling effect on the duration of the pre-bend delay.

The two sets of laser parameters do not have the same line energy, but doubling the scan speed while only slightly lowering the laser power results in nearly identical bend profiles. If the increased absorptivity from the ablation channel plays a critical role in determining the energy input, then the pre-bend regimes for the different laser line energies would differ. The sample that significantly ablated should have bent with significantly less required input energy than the sample that did not ablate. As both samples had the same bend profile regardless of the presence of ablation, there must be some other factor controlling the rate of heat buildup and the duration of the pre-bend regime such as the formation of an insulating oxide layer. Ablation during laser bending decreases the thickness of the sample in addition to increasing the absorptivity. As the material thins, the length of the thermal gradient decreases. Less material thermally expands and there is less surrounding material to resist the thermal expansion, reducing the stress that drives laser bending. It is likely that the decrease in thickness outweighs the increase in absorptivity during ablation resulting in a lower bend profile than a hypothetical sample with all the same parameters that did not ablate.

### **Creation of an Oxide Layer**

When bending in air, there is an increase in the final oxide thickness after all bending has finished, and increasing the number of passes or the temperature reached increases the oxide thickness [13]. When bending in different atmospheres, the oxygen content varies in addition to changing the cooling mechanism. Atmospheric pressure air has

the highest oxygen content while one atmosphere of argon has the lowest. Counter-intuitively, bending in the low air pressure environment resulted in the thickest oxide layer, as seen in Fig. 12. For these conditions, the amount of material removed by the sputter step before the XPS measurement was added to the measured thickness of the oxide layer, contributing to the wide data spread at thicker oxide layers. The exact cause of the greater oxidation at low pressures is currently unknown, but Fang et al. proposed a mechanism in which the rate of oxidation growth is faster at low pressures than at higher pressures [35]. In the similar process of oxidation of silicon at low pressures, the oxygen molecules have a higher kinetic energy due to the lower air pressure and thus diffuse deeper into the sample, increasing the thickness of the oxide layer [35]. It is possible that until at least medium vacuum levels are reached, in the range 10 mTorr as seen in Fig. 12, the thickness of the oxide layer will increase despite decreasing oxygen content.

The low pressure sample reaches a lower final bend angle than the ambient atmospheric one, likely because the less effective cooling leads to thermal saturation. The lower cooling rate combined with the high thermal conductivity of aluminum rapidly leads to an even heat distribution through the thickness of the sample. This decreases the thermal-induced stress, and results in the observed lower final bend angle. This thermal saturation is likely exacerbated by the increasing oxide layer thickness. DeCoster et al. found that increasing the thickness of the oxide layer increases the thermal resistivity until around 40 nm, at which point the insulating effect plateaus [36].

The thicker initial oxide layer created by the 450°C heat treatment is more thermally insulating than the thinner native oxide layer on the as-rolled specimens. Scott et al. states that thin oxide layers in the range of 1 to 10 nm have a thermal conductivity that is proportionally related the thickness [37]. From this, the difference in the duration of the pre-bend regime between the heat-treated and as-rolled low pressure samples can likely be described as a feature of the difference in oxide layer thickness and how quickly the oxide layer grows. The large variance in the pre-bend delay of the as-rolled samples bent in an argon environment is due to the fact that none of the samples left the pre-bend regime over the course of the bending process. The more insulating thicker oxide layer on the heat treated samples retains more heat within the sample, achieving a higher temperature quicker, resulting in the observed instantaneous bending and quicker kinetics for subsequent oxidation.

The rate of oxidation in aluminum increases with heat treatment temperature, heat treating at 400°C results in a much smaller oxide layer than heat treating at 450°C, even over short time periods [38, 39]. However, as seen in Fig. 4, there is no difference between bending a sample heat treated at 400°C or at 550°C, in atmospheric pressure air. In that condition, it is likely that sample was able to quickly oxidize and equalize the thickness of the oxide layer. Then the softening effect of the heat treatment became the driving force for the improvement in bend efficiency. In low pressure or argon environments, the as-rolled samples' initial oxide layer is too thin to serve as a strong thermal insulator. With the weaker thermal insulation, the samples take longer to reach the critical oxide formation temperature. As seen in Fig. 6, the heat treated samples reached a higher temperature quicker than the as-rolled samples in both low-oxygen environments. In the early stages of the pre-bend regime, there

is sufficient cooling in the low pressure environment counteracting the temperature buildup in the as-rolled samples. By the time the samples have reached the critical oxide formation temperature, the samples have thermally saturated and lost the thermal gradient driving bending, resulting in the observed greatly lengthened pre-bend delay and smaller final bend angle.

## Conclusion

This work examines the role of atmosphere in determining the bending behavior of the AA3003 Al alloy. In particular, it provides a preliminary analysis of and an explanation for the causes of the observed pre-bending regime during laser forming of aluminum alloys using low power lasers and describes parameters which can mitigate or control this effect:

- The duration of the pre-bending regime is controlled by the heating rate and how well the thermal gradient can be maintained during bending. Increasing the heating rate decreases the time to reach a threshold stress for bending, which decreases the duration of the pre-bending delay. For high rates of cooling a higher laser power is required to reach the thermal gradient necessary for bending and the thermal gradient cannot be maintained, elongating the duration of the pre-bend delay.
- The laser power is more important than the laser scan speed in determining the final bend angle and the duration of the pre-bend delay due to the high thermal conductivity of the material used. Materials with high thermal conductivity can achieve and equalize a thermal gradient within the interaction time of a single laser pass. The temperature and subsequently the bend angle is more controlled by the input energy during each pass than the time needed for the thermal gradient to form.
- Ablation does not play a significant role in determining the duration of the pre-bend regime nor the final bend angle. The increase in absorptivity due to the increasing ablation channel depth is outweighed by the decrease in material thickness decreasing the quantity of material constrained by the thermal gradient. The stress driving bending is reduced for samples that have significantly ablated.
- The starting thickness of the oxide layer and the ability to rapidly oxidize during the initial laser passes increases the maximum temperature and maintains the thermal gradient. Initially the surface oxide serves as a thermal insulator to help develop the thermal gradient and achieve the threshold temperature for bending. If the oxide continues to thicken, the effects of cooling on developing the thermal gradient are decreased and the sample thermally saturates, resulting in a lower rate of bending.
- The pre-bend regime can be shortened by prior oxidizing the sample before bending in low oxygen environment or by using a higher power laser to reduce the energy needed or time required to surpass the threshold stress for bending.

Further study is needed to characterize the evolution of the oxide layer during the laser forming process and to elucidate the role of the insulating oxide layer on the development of the thermal gradient and overall laser forming process.

## Supplementary information

Supplementary information depicting the complete data set is included.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s40516-026-00346-8>.

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**Data Availability** Data can be made available upon reasonable request.

**Materials Availability** Not applicable.

**Code Availability** Code is available at [github.com/MONSTERgroup/NINJA](https://github.com/MONSTERgroup/NINJA).

## Declarations

**Ethical Approval and consent to participate** Not applicable.

**Consent for publication** Not applicable.

**Competing interests** The authors declare no competing interests.

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